

KENYA NATIONAL EXAMINATIONS COUNCIL

KCPE 2005

SCIENCE

READ THESE INSTRUCTIONS CAREFULLY

Time:1 hour 40 minutes

1. You have been given this question booklet and a separate answer sheet. The question booklet contains 50 question
2. Do any necessary rough work in this booklet.
3. When you have chosen your answer, mark it on the **ANSWER SHEET**, not in this question booklet.
4. Use only an ordinary pencil.
5. Make sure that you have written on the answer sheet:

YOUR INDEX NUMBER

YOUR NAME

NAME OF YOUR SHOOOL

6. By drawing a dark line inside the correct numbered boxes mark your full Index Number(i.e School code number and the three-figured Candidate's number)in the grid near the top of the answer sheet.
7. Do not make any marks outside the boxes.
8. Keep the sheet as clean as possible and do not fold it.
9. For each of the question 1-50 four answers are given. The answers are lettered A,B,C and D. In each case only ONE of the four answers is correct. Choose the correct answer.
10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the letter have chosen is written.

Example

In the Question booklet.

4. Which one of the following components of blood is involved in the clotting after injury?

- A. Plasma
- B. White blood cells
- C. Red blood cells

D. Platelets

The correct answer is D

1 [A] [B] [C] [D] 14. [A] [B] [C] [D] 24. [A] [B] [C] [D] 34. [A] [B] [C] [D]

11. Your dark line MUST be within the box.

12. For each question ONLY ONE box is to be marked in each set of four boxes.

This question paper consists of 7 printed pages and 1 black page.

The Kenya National Examination Council, 2005.

1. Which one of the following sources of energy is renewable?
 - A. Kerosene.
 - B. Biogas from cow dung.
 - C. Coal.
 - D. Cooking crude oil.
2. Which one of the following is the **MAIN** reason why the inside of the solar drier is painted black?
 - A. To prevent rusting.
 - B. To make it last longer.
 - C. To make it absorb heat.
 - D. To decorate it.
3. The image formed in the pinhole camera is
 - A. upright and smaller than the object.
 - B. upside down and formed on the screen.
 - C. upside down and same size as object.
 - D. upright and formed on the screen.
4. In a box guitar high sound can be produced without changing the tension of strings by
 - A. plunking the string hard.
 - B. increasing the length of strings.
 - C. using thinner strings.
 - D. increasing the size of the box.
5. Below is an energy transformation:
Chemical- $\rightarrow$ electrical- $\rightarrow$ heat- $\rightarrow$ light.
In which one of the following does the energy transformation shown take place when used to produce light?
 - A. Geothermal.
 - B. Car battery.
 - C. Electric bulb.
 - D. Bicycle dynamo.
6. Flowers **P**, **Q**, **R** and **S** have the following characteristics:
P: *large and loosely held anthers.*
Q: *presence of nectar.*
R: *strong scent.*
S: *lighter grains.*
Which two flowers are wind pollinated?
 - A. **P** and **S**.
 - B. **Q** and **R**.
 - C. **P** and **R**.
 - D. **Q** and **S**.

7. A fish is able to move **EASILY** in water because it
- has fins.
 - has hard scale.
 - is streamlined.
 - has gills.
8. Which one of the following plants stores food in the stem?
- Carrot.
 - Irish potatoes.
 - Sweet potatoes.
 - Onion.

9. A food chain in grassland is shown below.

Grass-> grasshoppers->frogs->snakes-> secretary birds.

Which one of the following would most likely happen immediately all snakes are killed?

- Frogs would decrease in number.
 - Grasshoppers would increase in numbers.
 - The amount of grass would increase.
 - Secretary bird would decrease in numbers.
10. The diagram below represents a section of heart of mammal.

DIAGRAM

Which one of the following statement is **TRUE**?

- K** carries blood into the heart whereas **L** takes it out of the heart
 - K** has thin wall whereas **L** has thick wall
 - K** carries deoxygenated blood whereas **L** carries oxygenated blood
 - K** has valves whereas **L** has no valves
11. The following are activities which are carried out when comparing strength of acids in the fruit juices.
- Counting the number of drops of each fruit juice required to change the colour.*
 - Adding the different fruit juices drops by drop in each of the bottle tops.*
 - Addition equal amount of indicator in each of the bottle top.*
 - Putting equal amounts of wood as solution in to the different bottle tops.*
- Which one of the following is the **CORRECT** order of the activities?
- (iv) (iii) (ii) (i).
 - (iv) (ii) (iii) (i).
 - (iii) (i) (iv) (ii).
 - (ii) (iii) (i) (iv).

12. The diagram below represents a set up that can be used to demonstrate a certain property of air.

DIAGRAM

Which property of air is demonstrated by the set up?

- A. Air exerts pressure in all direction.
 - B. Air occupies space.
 - C. Parts of air is used in burning.
 - D. Air expands when heated.
13. Which one of the following will pollute water **ONLY**?
- A. Untreated water.
 - B. Used soil.
 - C. Farm chemical.
 - D. Factory wastes.
14. The **MAIN** reason why the government conserve wildlife is to have?
- A. meat from wild animals.
 - B. plants and animals products.
 - C. a balance of nature.
 - D. animals for exports.
15. When an object is thrown up in the air it falls because it
- A. is pulled by the earth.
 - B. exerts pressure.
 - C. is resisted by air.
 - D. is as heavy as air.
16. The diagram below represents a simple machine in use.

DIAGRAM

The machine is an example of

- A. an inclined plane.
 - B. a wheel and axle.
 - C. a lever.
 - D. a pulley.
17. The reason for using rollers when moving heavy objects on floor is to
- A. makes object lighter.
 - B. avoid damaging the floor.
 - C. increase speed of moving objects.
 - D. reduce force that opposes movements.

18. The diagram below represents a weather instruments constructed by a group of pupils who made a mistake while constructing it.

DIAGRAM

The mistake made during the construction was

- A. closing the mouth of the bottle.
 - B. using little amount of coloured water.
 - C. having the container with coloured water open.
 - D. having the scale reading from the bottom.
19. Which one of the following planets are in the **second, fifth, eighth**, position respectively from the sun?
- A. Uranus, Jupiter, Venus.
 - B. Mars, Saturn, Neptune.
 - C. Venus, Jupiter, Neptune.
 - D. Earth, Saturn, Uranus.
20. The function of plasma is to
- A. fight germs.
 - B. combine with oxygen.
 - C. help in clotting of blood.
 - D. transport food.
21. Which pair of characteristic is common to lizard, toad, hawk and platypus?
- A. Lay eggs, breath by lungs.
 - B. Body temperature varies, breathe by lungs.
 - C. Have scale, lay eggs.
 - D. Body temperature varies, have scale.
22. The table below shows characteristic of four different fruits **U, V W** and **X**

Fruit	Characteristics
U	Small and hairy
V	Long with lines of weakness
W	Brightly coloured with hard seed coat
X	Hollow and fibrous

Which one of the fruits is likely to be dispersal by exploding?

- A. **U**.
- B. **V**.
- C. **W**.
- D. **X**.

23. The diagram below represents a foot of a certain bird.

DIAGRAM

Which one of the following diagrams of beaks belong to the same bird

24. Which one of the following parts of a flower is **NOT CORRECTLY** matched with its functions?

Parts of a flower	Function
A. Stigma	produce pollen grains.
B. Pollen grains	fertilize ovules.
C. Style	supports stigma.
D. Filaments	support anthers.

25. A glass containing ice cubes was placed in an open. After some time the outer surface of the glass became wet. This is to show that

- A. air contains water vapour.
- B. ice melts to form water.
- C. air is a mixture of gases.
- D. air can be cooled.

26. Which one of the following is **NOT** required when finding the density of a regular solid?

- A. Spring balance.
- B. Ruler.
- C. Overflow can.
- D. String.

27. Which one of the following statements is about water is **TRUE**?

- A. Water mixes with all liquids.
- B. Water has definite size.
- C. Water has definite shape.
- D. Water is good conductor.

28. The **best** way of handling tin cans after using the contents is by

- A. Burying them deep in the soil.
- B. Smelting and moulding them into other products.
- C. Throwing them into pit latrine.
- D. Throwing them in river.

29. Which one of the following pairs of levers has position of load, fulcrum and effort as forearm?
- A. Fishing rod and pliers.
 - B. Spade and tongs.
 - C. Bottle top and nutcracker.
 - D. Scissors and claw hammer.
30. Which one of the following statements about wind sock is **TRUE**? A windsock
- A. measures speed of wind.
 - B. shows direction wind is blowing from.
 - C. shows the strength of the wind.
 - D. works the same way as wind vane.
31. Which one of the following **NEED NOT** be the same when comparing capillarity in different types of soil?
- A. Diameter of tubes.
 - B. Amount of each type of soil.
 - C. Time allowed for the experiment.
 - D. Amount of water used.
32. Which one of the following characteristics of sandy soil **DOES NOT** explain why water drains through easily?
- A. Rough texture.
 - B. Large particle.
 - C. Large air space.
 - D. Shiny particle.
33. Which one of the following drugs has similar effects to *mirra* (khat) when abused
- A. Heroin.
 - B. Tobacco.
 - C. Mandrax.
 - D. Excessive alcohol.
34. Which one of the following correctly shows the path taken by urine from the kidney?
- A. Ureter-> urethra->bladder.
 - B. Urethra-> ureter->bladder.
 - C. Ureter-> bladder->urethra.
 - D. Urethra-> bladder->ureter.
35. Which one of the following groups consists only plants with fibrous roots
- A. Cowpeas, maize, wheat.
 - B. Rice, millet, maize.
 - C. Carrot, millet, wheat.
 - D. Cowpeas, rice, carrot.

36. The component of air used in making plant food is
- A. 0.03%
 - B. 0.97%
 - C. 21%
 - D. 78%
37. Which one of the following lists consists of only drugs of abuse that are prohibited in Kenya?
- A. Khat, tobacco, cocain.
 - B. Cocaine, heroin, mandrax.
 - C. Bhang, beer, tobacco.
 - D. Cobblers' glue, khat, beer.
38. Which one of the following processes involves loss of heat to surroundings?
- A. Melts and freezing.
 - B. Evaporation and condensation.
 - C. Freezing and condensation.
 - D. Melting and evaporation.
39. The leaf of a cactus plant
- A. Reduces water loss.
 - B. Makes plants food.
 - C. Stores water.
 - D. Allows gases exchange.
40. Which one of the following **DOES NOT** affect floating and sinking of an object?
- A. Size.
 - B. Shape.
 - C. Density.
 - D. Weight.
41. Which one of the following soil component is **LEAST** affected when soil in a container is heated?
- A. Air.
 - B. Shape.
 - C. Density.
 - D. Weight.
42. Which one of the following insect **DOES NOT** undergo the stage of eggs, larva, pupa and adult
- A. Wasp.
 - B. Locus.
 - C. Bee.
 - D. Moth.

43. Which one of the following substances will form a salt and water only when mixed with an acid?
- A. Ash solution.
 - B. Chalk powder.
 - C. Clover juice.
 - D. Magadi soda.
44. Which one of the following pairs of substances will give the same colour when mixed with flower extract?
- A. Lemon juice and lime solution.
 - B. Aspirin solution and actal solution.
 - C. Lemon solution and aspirin solution.
 - D. Lime solution and actal solution.
45. Sifting, picking and sieving are some methods of separating mixtures. Which one of the following mixtures can be separated by all the three methods?
- A. Maize and beans.
 - B. Sand and rice.
 - C. Maize flour and small pieces of iron.
 - D. Sugar and tiny pieces of glass.
46. Which one of the following is **NOT** a reason why drugs should be stored in their properly labeled container? To avoid
- A. contamination.
 - B. getting expired.
 - C. getting spoiled.
 - D. wrong usage.
47. Which one of the following materials is magnetic?
- A. Aluminum.
 - B. Steel razor blade.
 - C. Silver coin.
 - D. Copper ring.
48. The following are steps followed by pupils when making a certain tool
- i) *Flattening one end of a nail.*
 - ii) *Fixing the flattened end into a piece of wood.*
 - iii) *Cutting off the head of the nail.*
 - iv) *Flattening and sharpening the exposed nail.*

The tool is

- A. Hammer.
- B. Chisel.
- C. Drill.
- D. saw.

49. The diagram below represents a loaded wheelbarrow.

DIAGRAM

The effort needed to push the wheelbarrow can be reduced by increasing the length of the part between

- A. P and R.
- B. Q and R.
- C. R and S.
- D. S and T.

50. The diagram below represents an arrangement of gear wheels.

DIAGRAM

When wheel **H** is rotated in clockwise direction

- A. Wheel **G** rotates three times anticlockwise and **J** twice anticlockwise.
- B. Wheel **G** rotates three times anticlockwise and **J** twice clockwise.
- C. Wheel **F** rotates once clockwise and **G** three times clockwise.
- D. Wheel **F** rotates once anticlockwise and **J** twice anticlockwise.